Walk calibration of BTL modules in testbeam:

Via Synchronization of MCP scope data with FEBd

CMS

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Discussion draft; not approved for public release.

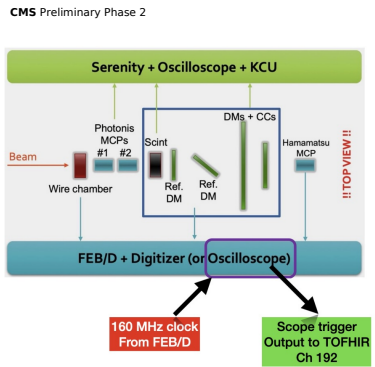
Abstract

Fixed-threshold timing depends on signal amplitude. An external microchannel-plate photomultiplier (MCP-PMT) provides a reference for measuring this time walk in BTL channels.

The MCP oscilloscope and BTL readout use independent clocks. Trigger matching, segment-wise clock alignment and LOWESS smoothing bring their times into a common coordinate. Quadratic fits in inverse charge then describe the walk.

For the same selected events, correction reduces the bar–MCP core width from 77.69 to 64.16 ps. This validates the correction under fixed selections; the width still includes detector, MCP and residual clock effects.

The reference and BTL readout must share a time coordinate



Problem

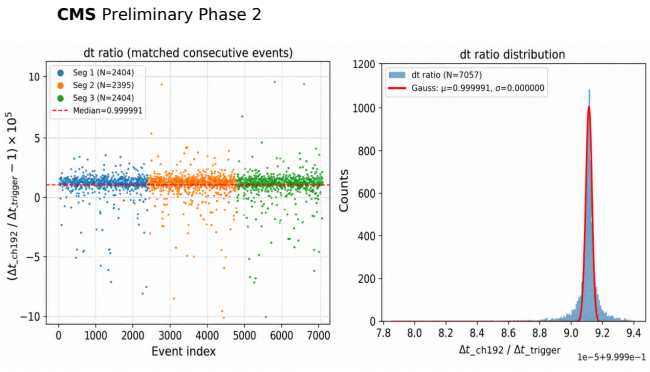
Smaller pulses cross a fixed threshold later. Measuring this delay requires an independent arrival-time reference.

Readout

The scope records the MCP pulse. Its trigger also reaches TOFHIR channel 192 in the FEBd readout.

September 2025 beam test. The highlighted lower path provides the reference-bar data. The trigger link identifies coincident records; the clocks still need alignment.

1. Match events using the intervals between triggers

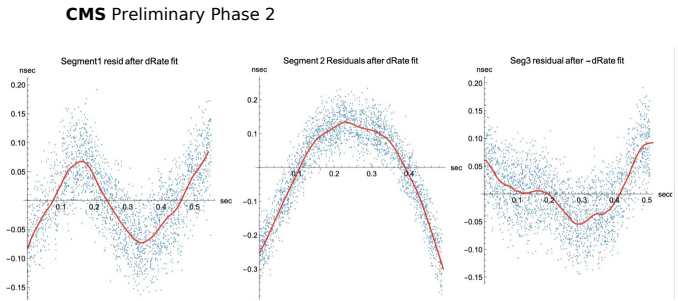


1. Compare consecutive trigger intervals in channel 192 and the scope.
2. Find consistent interval patterns.
3. Keep only unique event matches.

Matching identifies the same particle. It does not yet establish picosecond timing.

The correlated intervals and their ratio test event matching, not detector resolution. GitHub: trigger matching.

2. Correct clock-rate differences and smooth drift

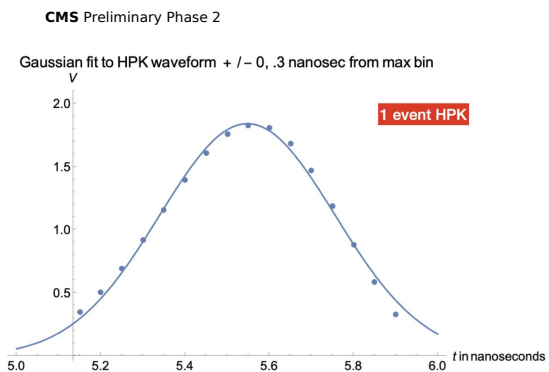


1. Split data at clock resets.
2. Fit a line between matched trigger times in each segment.
3. Fit the remaining drift with LOWESS.

Add the drift to the line prediction to obtain $g(u)$, the expected TOFHIR trigger time.

Points show residuals before LOWESS; red curves describe smooth drift. Reject groups with pre-LOWESS residual scale $1.4826\,\mathrm{MAD} > 1$ ns. GitHub: clock reconstruction.

3. Use the MCP pulse to define the reference time



Fit the MCP waveform to estimate its pulse time.
Within the scope waveform, define

$$\delta_{\mathrm{MCP}} = t_{\mathrm{peak}} - t_{\mathrm{trigger}}.$$

For SiPM channel i, use

$$\Delta t_i = t_i - g(u) - \delta_{\mathrm{MCP}}.$$

The charge-dependent part of this difference is the walk.

The waveform and fitted template determine the MCP pulse time. u is the scope-trigger timestamp. A fixed cable/geometry offset is irrelevant to the shape; clock-rate conversion over the short waveform interval is neglected.

4. Build charge profiles from a common selected sample

1. In run 4399, require exactly five TOFHIR hits and a unique MCP match: the target upstream pair, downstream channels 234 and 243, and channel 192.
2. Require waveform and clock quality, finite positive charge and slope-quality quantities. Keep charge at or above the spectrum's left half-maximum boundary.
3. Form 20 uniform $1/Q$ bins. Clip timing outliers at $5(1.4826\,\mathrm{MAD})$ from each bin's median; require at least five entries.
4. Fit the bin means with weights $1/\mathrm{SEM}^2$:

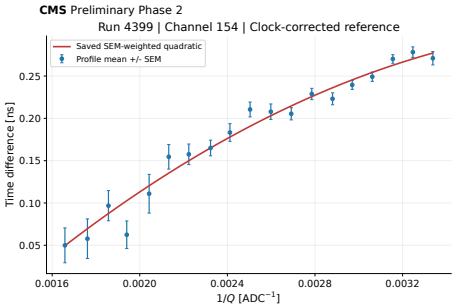
$$W_i(Q) = a_i + b_i/Q + c_i/Q^2.$$

Q is the measured ADC charge proxy, not energy in MeV. SEM denotes the standard error of the mean.

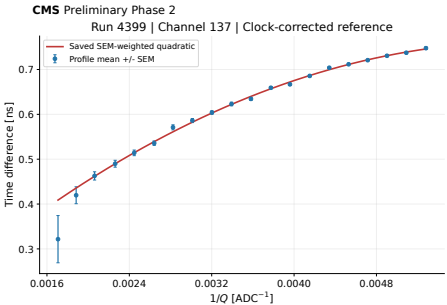
Reference pairs: bar 7 = channels 133 and 154; bar 8 = channels 137 and 150. GitHub: selected reference sample.

Each channel needs its own walk calibration

Bar 7: channel 154

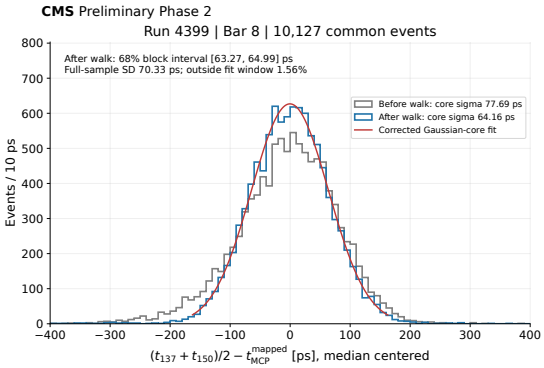


Bar 8: channel 137



Run 4399 examples. Points show mean time differences with SEM; red curves are SEM-weighted quadratics in $1/Q$. Only populated charge support is shown. The $+26$ ns display offset does not change the walk shape. The full note also shows channels 133 and 150.

5. Time difference before and after walk correction



10127 common events
Subtract each channel's fitted walk, then form the bar-average time minus MCP time.
Core width

$$77.69 \longrightarrow 64.16 \text{ ps}$$

Corrected full-sample SD: 70.33 ps.
This is not the intrinsic bar resolution.

Fixed clock mapping; a truncated-normal core fit uses median $\pm 2.5(1.4826\,\mathrm{MAD})$. The corrected-width 16–84% interval is [63.27, 64.99] ps from 400 file/segment-block bootstraps with fixed calibrations.

Use synchronized MCP timing to calibrate the walk shape

1. Match events and reject poor clock groups.
2. Reconstruct the reference time with a line plus LOWESS drift.
3. Fit each channel's timing profile versus inverse charge.
4. Subtract the fitted walk within its measured charge range.

Result: the same-event bar–MCP distribution becomes narrower.

Scope: quoted intervals are conditional statistics. Position, correlations, residual clock offsets and calibration systematics remain outside them.

The full MCP reference note retains the detailed selections, bibliography and uncertainty discussion. GitHub: timing correction; GitHub: reference selection.

References

[1] GitHub repository, `https://github.com/De-Cristo/LecroyTranslator4CMSTimingDetector`.

[2] Presentation by the Saclay group. `https://indico.cern.ch/event/1619706/`

[3] "Timing Performance of a Micro-Channel-Plate Photomultiplier Tube", `https://arxiv.org/pdf/1909.12604`

[4] CMS Collaboration, "A MIP Timing Detector for the CMS Phase-2 Upgrade". Technical Report CERN-LHCC-2019-003. CMS-TDR-020, CERN, Geneva, 2019 and "A High-Granularity Timing Detector for the ATLAS Phase-II Upgrade", CERN-LHCC-2020-007; ATLAS-TDR-031

[5] S. White, Nov. 2025 BTL TB Analysis mtg presentation, `https://indico.cern.ch/event/1614521/contributions/6803723/attachments/3178820/5922530/5vs10vs20GSa_s.pdf`

[6] W. R. Leo, "Techniques For Nuclear and Particle Physics Experiments", p.326, Springer-Verlag Berlin Heidelberg.

[7] S. White and A. Boletti, "Amplitude Walk in Fast Timing: The Role of Dual Thresholds" 2026 JINST V21 P02001.

[8] W. S. Cleveland, "Robust Locally Weighted Regression and Regression Scatterplots", J. Amer. Statist. Assoc. **74** (1979) 829.